

164 FEET

THE DISTANCE TRAVELED BY THE ADER ÉOLE



Clément Ader's flying machine was the first piloted, heavier-than-air machine to take off, literally under its own steam, on October 9, 1890.

ON OCTOBER 1, THE US CONGRESS PASSED AN ACT that founded the **Yosemite National Park in California**. This brought the park, which had existed since 1872, under federal control.

The process of **meiosis**—a stage in cell division responsible for the production of gametes—had first been described by German biologist Oscar Hertwig (1849–1922) during his study of sea urchin eggs in 1876. The full significance of meiosis in reproduction and inheritance was appreciated only with the work of German biologist **August Weismann** (1834–1914) in 1890. He realized that two cell divisions are necessary to transform one diploid cell (with two sets of chromosomes) into four haploid cells (with one set of chromosomes each) to maintain



Robert Koch

This lithograph—copied from an 1890s photograph—shows the German bacteriologist working on the Rinderpest virus in his laboratory.

tetanus, into an animal caused its blood to **produce antibodies** that gave immunity against the disease. This provided a practical counterpoint to German physician Robert Koch's (1843–1910) theories about the relationship between microbes and disease, which were published in the same year. German bacteriologist Friedrich Löffler (1852–1915) had worked with Koch to develop these ideas.

Arguably, the most important book in the history of psychology, ***The Principles of Psychology***, by American philosopher and psychologist **William James**

(1842–1910), was published in 1890. Comprising two volumes and 1,200 pages, the book took James 12 years to write and covered everything in the field known at the time.

this feat 13 years before the Wrights. His machine—the Ader Éole—had a batlike design and a wingspan of 46 ft (14 m). It was powered by a lightweight, four-cylinder steam engine with

“AN AIRPLANE-CARRYING VESSEL IS INDISPENSABLE... IT WILL LOOK LIKE A LANDING FIELD.”

Clément Ader, French inventor, from *L'Aviation Militaire*, 1909

Contrary to popular belief, the Wright brothers did not make the **first manned flight of a heavier-than-air flying machine**. This accolade goes to the French inventor **Clément Ader** (1841–1926), who achieved

20 horsepower—that weighed only 112 pounds (51 kg). On October 9, the aircraft took off—with Ader on board—and reached a height of 8 in (20 cm). It flew uncontrolled for roughly 165 ft (50 m).

MEIOSIS

Meiosis, the division of cells for sexual reproduction, produces gametes, such as sperm and egg cells. Chromosomes from two parents undergo “recombination,” which shuffles the genes to produce a different genetic combination in each gamete. Meiosis produces four genetically unique cells, each with a single set of chromosomes.

